Spice Community Archetype Report

Operation Spice Route Compass

Resolving Aroma Drift with Hierarchical and K-Means Clustering

Prepared for the Global Spice Heritage Foundation Board and the Spice Producer Alliance

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Executive Summary

Operation Spice Route Compass addresses the "Aroma Drift" problem identified by the Global Spice Heritage Foundation. Spice-farming communities show differences in quality ratings, market prices, and export growth, even though support programs have been applied across the wider production network. Hierarchical clustering and k-means clustering were applied to the `silk_route_spice_data.csv` dataset to identify data-driven community archetypes.

The analysis covered 300 communities across 20 numeric variables spanning environmental conditions, cultivation practices, infrastructure, socio-economic indicators, and market access. Categorical identifiers were excluded from distance calculations and used only to interpret cluster assignments. Missing values were replaced with variable medians, all numeric variables were standardised, and k-means was run with 25 random starts to ensure stable results.

A four-cluster solution was selected based on the dendrogram structure, the elbow plot, and the highest average silhouette score (0.367) among all tested values of k. The four archetypes are:

- Cluster 1: Digital Market Connectors
- Cluster 2: Lowland Volume Producers
- Cluster 3: High-Altitude Artisan Specialists
- Cluster 4: Infrastructure-Constrained Traditional Communities

The results indicate that a single support program is unlikely to address the structural differences between these groups. Tailored recommendations for each archetype are presented in the Targeted Strategies section.

1 Introduction

This report presents a cluster analysis of spice-growing communities along the Silk Route. The dataset combines environmental, production, infrastructure, social, and market-related variables for 300 communities. The aim is to identify groups of communities with similar characteristics and interpret these groups in operational terms.

The analysis follows a local-perspective scenario. Spice production and trade remain important in many South Asian regions, but communities differ in natural conditions, production practices, infrastructure access, digital readiness, and export potential. Clustering can make these differences visible by grouping communities with similar strengths and constraints. These groups can support more targeted decisions on training, infrastructure improvement, digital sales support, cooperative development, and export planning.

The objectives of this analysis were:

- Preprocess and clean the dataset through column selection, median imputation, and numerical scaling.
- Evaluate cluster structures using Ward's hierarchical clustering, the elbow method, and silhouette validation.
- Partition the 300 observations using the k-means algorithm and select an appropriate number of clusters.
- Interpret the resulting community profiles as practical archetypes.
- Derive targeted support strategies for each archetype.

2 Data Preparation and Preprocessing

The dataset contains 300 observations and 23 variables, with one row representing one farming community. Three variables — community ID, region, and dominant spice type — are categorical identifiers and were excluded from the distance-based clustering procedure. The remaining 20 numeric variables cover environmental conditions, cultivation practices, infrastructure, socio-economic factors, and market readiness.

Missing values were identified in four variables: organic certification status (12), artisan training hours (23), cooperative membership rate (14), and online sales channel usage (27). Each was replaced with the column median [1]; no missing values remained after imputation.

Outliers were screened using the 1.5 IQR rule. Road infrastructure quality had 22 flagged values; electricity access had 6; soil organic matter and literacy each had 1. These observations were retained because they reflect genuine variation in community conditions — limited electricity or road access, for example — that is directly relevant to the Aroma Drift problem.

All 20 numeric variables were standardised to zero mean and unit variance before clustering. Standardisation was necessary because the variables span incompatible units and ranges, from altitude in metres to export potential on a 1–10 scale; without it, variables with large absolute ranges would disproportionately influence the distance calculation [2].

3 Clustering Methodology

All analyses were performed in R [3]. The cluster analysis methodology follows the framework described in Kaufman and Rousseeuw [4].

3.1 Hierarchical Clustering

Euclidean distance and Ward's `ward.D2` linkage were applied [5, 6]. Ward's method minimises within-cluster variance at each merge step, producing compact and internally homogeneous groups. The dendrogram shows several large branch separations before finer local splits. A cut at four clusters yields interpretable groups without over-fragmenting the data.

3.2 K-Means Clustering

K-means [7, 8] was applied as the primary segmentation method, with 25 random starts (`nstart = 25`) per run to reduce sensitivity to initialisation. The elbow method [9] was calculated for 1 to 10 clusters, and average silhouette scores [10] were calculated for 2 to 10 clusters:

Clusters	2	3	4	5	6	7	8	9	10
Silhouette	0.291	0.355	0.367	0.316	0.253	0.171	0.098	0.102	0.098

The silhouette score peaks at four clusters (0.367), and the elbow plot shows an inflection at the same point. The four-cluster solution was therefore selected, with hierarchical clustering used as exploratory validation. The complete R implementation is provided in Appendix A.

4 Cluster Archetypes

Cluster	Archetype	Count	Dominant region	Dominant spice	Export
1	Digital Market Connectors	75	Western Silk Route	Cinnamon	5.69
2	Lowland Volume Producers	90	Eastern Silk Route	Turmeric	3.23
3	High-Altitude Artisan Specialists	75	Western Silk Route	Saffron	8.20
4	Infrastructure-Constrained Traditional Communities	60	Central Silk Route	Mixed Herbs	2.30

4.1 Key Variable Profile

The table below summarizes cluster means for the nine most distinguishing variables. The complete numeric breakdown is available in `outputs/tables/full_cluster_profile.csv`.

Variable	C1	C2	C3	C4
Altitude (m)	802.75	284.29	2524.00	1505.38
Rainfall (mm)	962.03	1199.31	695.45	803.63
Soil organic matter (%)	70.20	60.03	84.83	75.69
Water reliability index	78.27	68.79	91.13	61.10
Traditional methods score	6.85	5.23	8.76	8.22
Storage capacity index	72.84	56.50	79.31	47.78
Export potential score	5.69	3.23	8.20	2.30
Literacy rate (%)	74.84	64.97	79.12	51.05
Online sales usage (%)	70.23	16.31	29.41	13.07

4.2 Archetype Interpretation

Cluster 1: Digital Market Connectors. These communities combine high online sales usage (70%), strong electricity access, modern processing technology, and high youth engagement. Export potential is moderate. The profile suggests communities already embedded in digital commerce and well placed to develop traceable, origin-branded supply chains.

Cluster 2: Lowland Volume Producers. Located at low altitudes with high annual rainfall and the shortest average distance to major markets, these communities have the largest farm sizes and the strongest local demand. Organic certification and online sales are low. The profile is consistent with high-volume, near-market supply, constrained mainly by post-harvest consistency and storage capacity.

Cluster 3: High-Altitude Artisan Specialists. These communities operate at the highest altitudes, with strong water reliability, the highest soil organic matter, and the highest rates of organic certification and cooperative membership. Export potential is the highest of the four clusters. Distance to major markets is a constraint, but the environmental and social profile supports a premium, origin-based positioning.

Cluster 4: Infrastructure-Constrained Traditional Communities. These communities show strong adherence to traditional methods but face the weakest infrastructure across electricity, roads, and storage. Literacy and youth engagement are the lowest of any cluster, and distance to markets is the greatest. The low market performance of this group appears to reflect structural access barriers rather than a lack of production capability.

5 Targeted Strategies

Cluster 1: Digital Market Connectors. The priority is to convert existing digital access into consistent quality signals. Support should cover digital merchandising, export documentation, and traceability systems. Quality control should introduce lot-level grading and repeatable post-harvest checklists. Market development should target premium online channels, specialty retailers, and diaspora buyers, with branded cinnamon and origin-based blends as core products.

Cluster 2: Lowland Volume Producers. The priority is supply-chain reliability. Support should address drying, sorting, storage, and cooperative aggregation. Quality control should standardise moisture limits, contamination thresholds, and grading before produce enters bulk channels. Market development should prioritise domestic wholesale and institutional buyers as the initial route, with export expansion deferred until quality consistency is established.

Cluster 3: High-Altitude Artisan Specialists. The priority is to formalise the existing quality profile without weakening traditional expertise. Support should cover export packaging, certification renewal, and premium buyer-readiness training. Quality control should apply small-batch grading, organic certification audits, and sensory evaluation. Market development should target specialty importers, fair-trade channels, and gourmet retailers, with saffron, cardamom, and limited-harvest editions as flagship products.

Cluster 4: Infrastructure-Constrained Traditional Communities. The priority is foundational access. Support should address electricity reliability, road access, storage repair, and literacy-linked training, alongside cooperative strengthening and youth apprenticeship programmes. Quality control should remain simple and field-applicable: drying calendars, visual grade cards, and shared storage protocols. Market development should begin through cooperative collection points and foundation-managed buyer links rather than direct online selling.

6 Figures

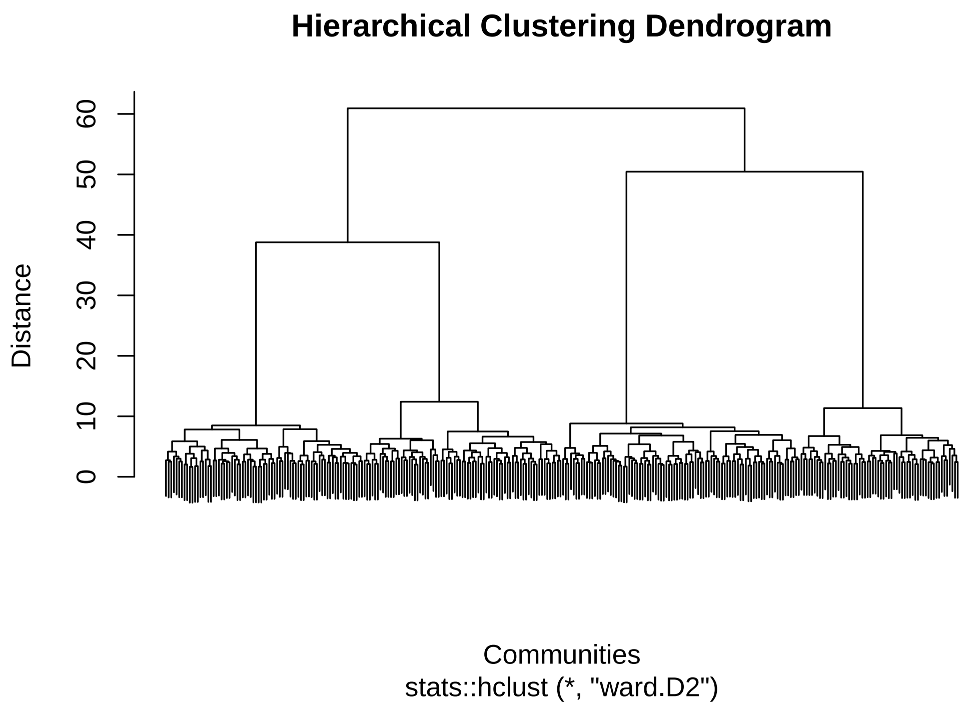

Figure 1: Hierarchical clustering dendrogram using Euclidean distance and Ward's method.

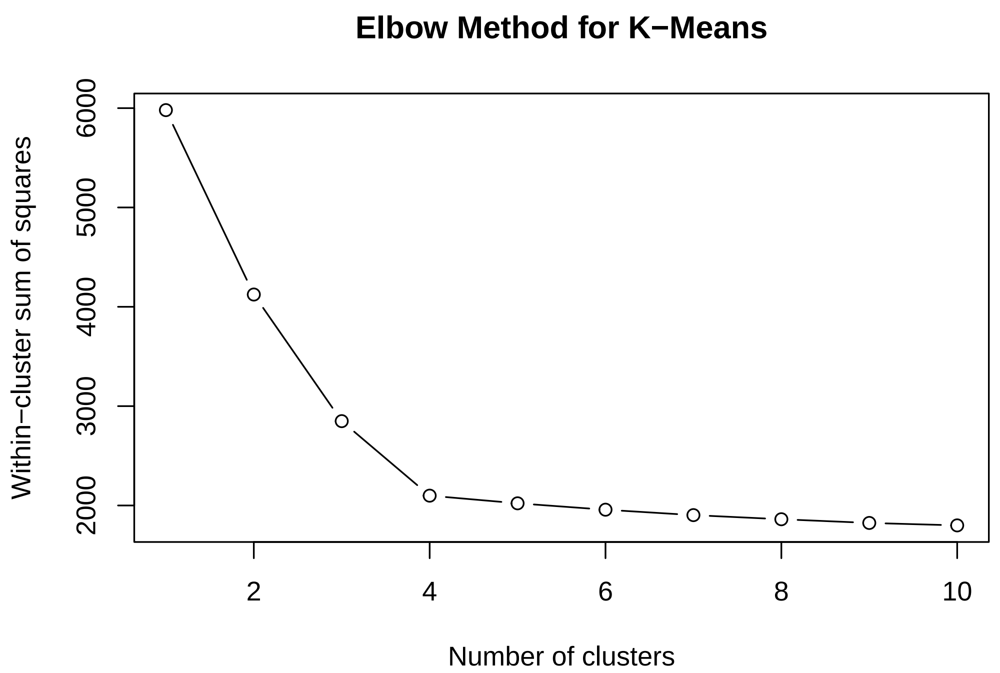

Figure 2: Elbow method for selecting the number of k-means clusters.

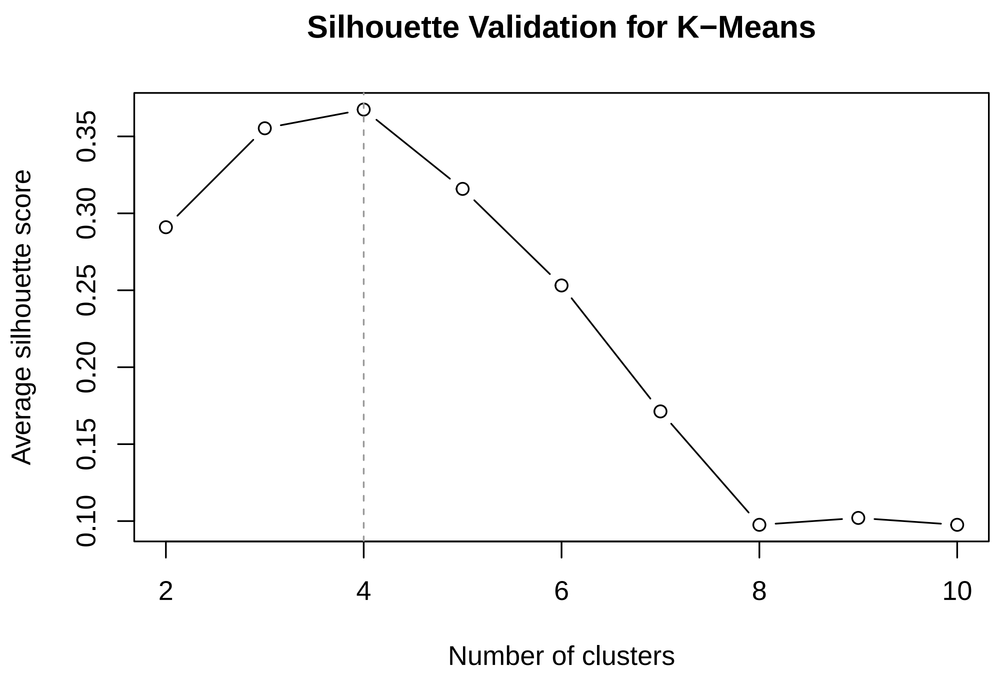

Figure 3: Average silhouette score by number of k-means clusters.

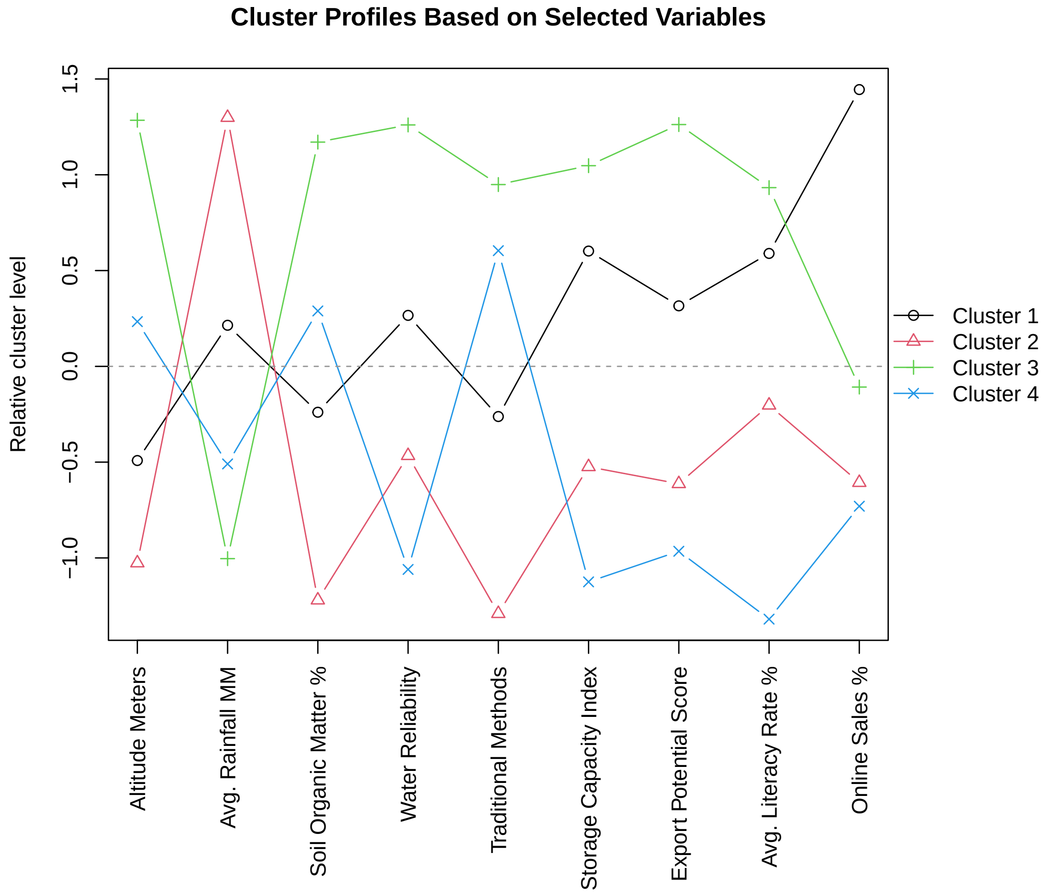

Figure 4: Profile comparison of the four final archetypes using selected distinguishing variables.

7 Conclusion

The four archetypes indicate that Aroma Drift is unlikely to be resolved by a single support program. The communities differ structurally in altitude, infrastructure quality, cooperative strength, digital market access, and export readiness. These differences require differentiated responses.

The clusters are most useful as four distinct intervention tracks: digital trade and traceability for Digital Market Connectors, post-harvest quality systems for Lowland Volume Producers, premium origin development for High-Altitude Artisan Specialists, and foundational access investment for Infrastructure-Constrained Traditional Communities. Applied consistently, this segmentation can support sustainable livelihoods and the long-term coherence of the Silk Route Spices brand.

8 References

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Appendix A: R Source Code

The analysis is implemented in two R source files submitted alongside this report. The complete project source code is publicly available at:

`https://github.com/gajalakshmiashwin/Spice-Community-Cluster-Analysis`

Appendix A.1 `main.R`

This script runs the complete analysis pipeline in sequence. It loads the dataset from `data/silk_route_spice_data.csv`, cleans missing values, scales the numeric variables, performs hierarchical clustering and k-means clustering, and writes all outputs. Figures are saved to `outputs/figures/` and data tables to `outputs/tables/`. The script is executed from the project root directory using:

```
Rscript main.R
```

Appendix A.2 `R/` — Helper Functions

The helper functions are organised into nine single-responsibility files, each sourced by `main.R` in pipeline order:

- `defaults.R` — shared default constant functions used across all stages
- `load_community_data.R` — `load_community_data()`, `understand_community_data()`, `count_missing_values()`
- `screen_outliers.R` — `summarise_numeric_outliers()`
- `preprocess_community_data.R` — `clean_community_data()`, `prepare_community_cluster_data()`, `scale_community_cluster_data()`
- `cluster_hierarchical.R` — `run_hierarchical_clustering()`
- `cluster_kmeans.R` — `run_k_means_clustering()`, `calculate_cluster_elbow_wss()`, `calculate_silhouette_scores()`
- `characterise_archetypes.R` — `calculate_full_cluster_profile()`, `calculate_cluster_sizes()`, `calculate_dominant_cluster_categories()`, `add_cluster_archetype_names()`, `calculate_cluster_profile_means()`
- `pipeline_utils.R` — assembly helpers (`assign_clusters()`, `build_cluster_summary()`), output path configuration, and export wrappers (`export_figures()`, `export_tables()`)
- `visualise_archetypes.R` — `plot_community_dendrogram()`, `plot_cluster_elbow()`, `plot_silhouette_scores()`, `plot_cluster_profile()`